*Computer Vision · Generative AI · Biometric Verification*

# Mask-Aware Multi-Reference Fusion

## with Verifier-Aware Sampling

*for Identity-Preserving Facial Inpainting*


TEAM MEMBERS

**Shaik Rehaan Siddiq**
23BCE8750

**Velaga Sasank**
23BCE8707

**Palivela Abhichand**
23BCE9968

**Mahendra Sidhu Chagantipati**
23BCE9056

PROJECT GUIDE

**Prof. Beebi Naseeba · VIT-AP University**


### Research Focus

- **Mask-Aware Reference Scoring**
- **Regional Identity-Token Fusion**
- **Quality-Constrained Biometric Reranking**
- **Severe-Occlusion Evaluation Protocol**

# Introduction — Why Plausible Completion Is Not Enough

## THE CORE PROBLEM

- Facial inpainting reconstructs missing or occluded facial regions in an image.
- **A perceptually realistic face can represent the wrong person — realism ≠ identity fidelity.**
- Standard benchmarks (SSIM, PSNR, FID) do not penalise identity drift.
- Under severe occlusion (>50% mask), discriminative identity cues vanish from the input.
- Biometric systems demand reconstructed faces pass face-verification — not just look good.

## TASK TAXONOMY

**Inpainting:** Fills missing/masked regions using surrounding context.

**Restoration:** Removes degradation (blur, noise) from an existing complete image.

**Editing:** Semantically alters attributes (age, expression, hair) of a face.

**Swapping:** Replaces the full identity of one face with another — different person.

*→ This work targets inpainting. The reconstructed region must match the subject's own identity.*

# Research Question & Research Gap

**PRIMARY RESEARCH QUESTION**

**Can region-specific evidence from multiple identity references improve biometric fidelity under severe facial occlusion without sacrificing realism?**

*Secondary: Can natural-language guidance provide controllable attribute correction during reconstruction?*

**RESEARCH GAP — GLOBAL REFERENCE USE IS INSUFFICIENT**

**Core gap:** Existing methods use reference identity information globally, even though different references contain unequal evidence for the specific facial region being reconstructed.

**Implication:** A reference showing a different pose may perfectly capture the ear but nothing about the nose — applying it globally dilutes the signal.

**Missing:** No existing method scores references conditioned on mask geometry before fusing identity tokens.

**Testable:** Identity fidelity should improve as reference relevance to the masked region increases — directly measurable.

# Research Objectives

## O1 — RETRIEVE

**Score references using pose, quality, visibility, and overlap with the missing facial region.**

- Estimate facial pose (yaw, pitch) for each reference.
- Compute face-quality score (sharpness, illumination, occlusion rate).
- Measure landmark visibility within the masked region.
- Score geometric overlap of each reference with the mask.
- Rank references; select top-K per region-of-interest.

## O2 — FUSE

**Construct spatial identity tokens from the most informative reference regions.**

- Extract patch-level identity features aligned to landmarks.
- Apply mask-conditioned cross-attention to weight regional tokens.
- Aggregate multi-reference tokens with a learned weighted sum.
- Inject fused identity tokens into the diffusion conditioning path.

## O3 — VERIFY

**Rerank or regenerate candidates using identity, realism, and structural constraints.**

- Generate K candidates from the diffusion model.
- Score each on identity similarity (held-out face matcher).
- Score on realism (NIQE / BRISQUE) and structural fidelity.
- Select candidate maximising the joint quality-identity score.
- Discard unrealistic images even if matcher similarity is high.

*Note: Text-guided attribute arbitration and inference efficiency are deferred to future work / optional experiments.*

# Literature Survey - Part 1 (Papers 1-4)

| Paper Title | Year | Method | Advantages | Limitations |
|---|---|---|---|---|
| PATMAT: Person Aware Tuning of Mask-Aware Transformer for Face Inpainting | 2023 | Person-Aware Tuning with Mask-Aware Transformer using multiple reference images. | High-quality face reconstruction with improved identity preservation. | Requires many reference images for tuning; no diffusion stage or explicit post-restoration verification. |
| Facial Feature Priors Guided Blind Face Inpainting | 2024 | Facial feature priors guide automatic blind face inpainting. | Automatically identifies degraded regions and restores facial features. | No explicit multi-reference identity conditioning; limited identity preservation under severe masks. |
| Masked Face Inpainting Using Denoising Diffusion Probabilistic Models | 2024 | DDPM-based inpainting with U-Net and Fast Fourier Convolution for global context. | High-quality coherent restoration with reduced edge blurring. | Computationally demanding; lacks multi-reference identity and explicit landmark guidance. |

# Literature Survey - Part 2 (Papers 5-8)

| Paper Title | Year | Method | Advantages | Limitations |
|---|---|---|---|---|
| Toward Real-World Blind Face Restoration With Generative Diffusion Prior | 2024 | BFRffusion uses pretrained Stable Diffusion with low-quality face feature guidance. | Photorealistic restoration and detailed reconstruction for real-world degradation. | Primarily single-image restoration; lacks explicit multi-reference identity conditioning. |
| Identity-Preserving Diffusion for Face Restoration | 2025 | LPIP-Diff models local patch relationships for identity-preserving diffusion restoration. | Improves restoration quality while maintaining identity consistency. | Does not explicitly use multi-reference feature fusion or post-restoration identity verification. |
| InpaintFormer: Prompt-guided High-Quality Face Inpainting with Mask-Aware Self-Attention | 2025 | CLIP-guided feature modulation with degradation prediction and mask-aware self-attention. | Combines semantic prompts with mask-aware attention for controlled face inpainting. | Does not explicitly integrate multi-reference identity encoding, landmark guidance, or post-reconstruction identity verification. |

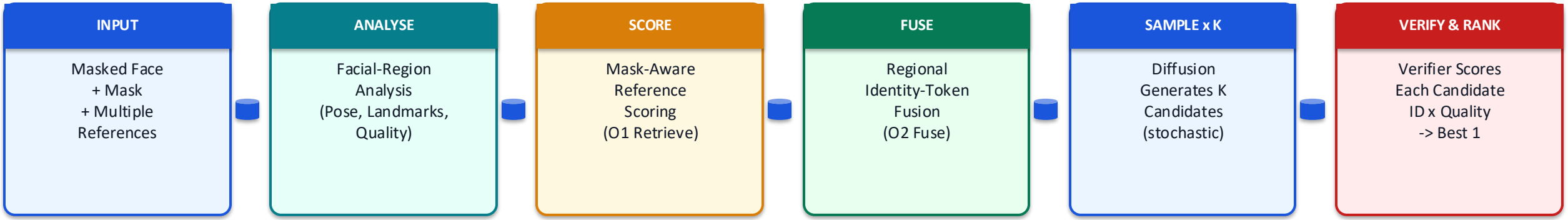
# Literature Survey - Part 3 (Papers 9-12)

| Paper Title | Year | Method | Advantages | Limitations |
|---|---|---|---|---|
| Towards Authentic Face Restoration with Iterative Diffusion Models and Beyond | 2023 | Iterative learned face restoration (IDM) system using progressive denoising steps. | Iterative refinement produces high authenticity compared to GAN-based models. | Iterative inference is computationally slow; single degraded image input without reference constraints. |
| Diffuse and Restore: A Region-Adaptive Diffusion Model for Identity-Preserving Blind Face Restoration | 2024 | Region-adaptive diffusion framework employing an identity-preserving conditioner. | Excellent perceptual quality and accurate identity preservation on face regions. | Relies on single-image restoration; does not support multi-reference feature fusion. |
| PFStorer: Personalized Face Restoration and Super-Resolution | 2024 | Personalization blocks integrated with generative priors to target specific user identities. | Outstanding fidelity to target individual's features and structural layout consistency. | Requires fine-tuning blocks per subject or access to target personal anchor codes. |

# Literature Survey - Part 4 (Papers 13-16)

| Paper Title | Year | Method | Advantages | Limitations |
|---|---|---|---|---|
| WaveFace: Authentic Face Restoration with Efficient Frequency Recovery | 2024 | Decomposes images into frequency components; restores low-freq via diffusion and high-freq via unified net. | Fast (single-step high-frequency restoration) and visually authentic details. | Frequency-split model lacks explicit multi-reference identity conditioning or verification. |
| OSDFace: One-Step Diffusion Model for Face Restoration | 2025 | One-step diffusion model utilizing a Visual Representation Embedder (VRE) and facial identity loss. | Extremely fast (one-step inference) with high realism and identity consistency. | Trained primarily for low-quality restoration rather than large-mask inpainting; lacks multi-ref fusion. |
| Robust ID-Specific Face Restoration via Alignment Learning | 2025 | Content injection and identity injection modules with multi-reference alignment learning (RIDFR). | Suppresses identity-irrelevant semantic noise (expression, pose, makeup) during recovery. | Designed for restoration of low-quality images rather than inpainting; lacks post-restoration verification. |

# Proposed Architecture - Mask-Aware Multi-Reference Fusion Pipeline




**VERIFIER-AWARE SAMPLING - Mechanism (Inference-Time Reranking, NOT a Training Loss)**

At inference: diffusion samples K independent outputs (K=8-16, stochastic). A held-out ArcFace-R100 matcher + NIQE quality estimator score each candidate. Final output = argmax( lambda*ID_similarity + (1-lambda)*quality ). The verifier never receives gradients - it is a test-time reranker only.

## KEY DESIGN DECISIONS

* **Mask-conditioned scoring (O1):** References scored by geometric overlap with mask - not global quality. Top-K selected per region.
* **Regional token fusion (O2):** Patch-level identity features weighted by mask-region relevance via cross-attention; injected into UNet.
* **Verifier-aware sampling (O3) = test-time reranker:** K candidates sampled; held-out verifier + quality score select best. No gradient through verifier.

# Novelty & Contributions

## Mask-aware identity evidence retrieval and verifier-aware candidate selection

### C1

**Mask-Conditioned Reference Scoring**

A scoring function that evaluates each reference for its geometric overlap with the occluded region, conditioning on mask geometry rather than treating all references equally.

### C2

**Regional Identity-Token Fusion**

Patch-level identity features aligned to landmarks, combined via mask-conditioned cross-attention to produce spatially-aware identity tokens for diffusion conditioning.

### C3

**Quality-Constrained Biometric Reranking**

Joint scoring by identity fidelity AND perceptual quality, preventing selection of unrealistic images based solely on matcher similarity.

### C4

**Severe-Occlusion Evaluation Protocol**

Systematic benchmark at four mask-severity levels (25%, 50%, 75%, full-face) with ablations and evaluation using a held-out face matcher.

*Note: We do not claim guaranteed identity preservation. The system maximises identity fidelity under perceptual quality constraints.*

# Software & Hardware Specifications

## SOFTWARE SPECIFICATIONS

- **Programming Language:** Python
- **Deep Learning Framework:** PyTorch
- **Image Processing:** OpenCV, NumPy
- **Model Architectures:** Hugging Face Diffusers & Transformers
- **Biometric Verification:** Pretrained face recognition / face embedding frameworks (e.g., ArcFace, Facenet)
- **Development Environment:** VS Code / Jupyter Notebooks, CUDA Toolkit

## HARDWARE REQUIREMENTS

- **Graphics Processing Unit:** NVIDIA CUDA-compatible GPU
- **Video Memory (VRAM):** 8–12 GB VRAM targeted for local experimentation, adapter tuning, and inference
- **System RAM:** 16 GB+ RAM
- **Storage:** High-speed SSD for dataset loading and model checkpoints

*Implementation Note: Pretrained Transformer/diffusion modules will be adapted or fine-tuned (e.g., via adapters or LoRA) to fit within student-accessible GPU resource constraints (8–12 GB VRAM).*

# Datasets & Experimental Pipeline

**CelebA-HQ**
**(1024x1024 high-res face images, provides identity metadata to build multi-reference groups)**

**CelebAMask-HQ**
**(Provides 19-class segmentation masks for eyes, nose, mouth, skin, hair, and accessories)**

**Flickr-Faces-HQ (FFHQ)**
**(70k high-quality faces with variations in age, accessories, and background; used for general evaluation)**

## EXPERIMENTAL PIPELINE & REFERENCE CONSTRUCTION

1. Select identities with sufficient suitable images for multi-reference pairing.
2. Select one image as the ground-truth target and up to three other images as reference inputs.
3. Apply synthetic masks/occlusions to the target to produce the damaged input.
4. Use the references to provide identity information under variations in pose, expression, lighting, and viewpoint.
5. Evaluate reconstructed output against the original ground-truth target.

# Evaluation Plan - Identity Fidelity vs. Mask Severity

PRIMARY EXPERIMENT

## Identity Fidelity vs. Mask Severity

Four severity levels: 25% | 50% | 75% | Full-face

*Full-face: when 100% is masked, identity evidence comes entirely from the reference set, directly testing retrieval and fusion in isolation.*

### Metrics & Protocol

* **Identity (ArcFace-R100):** Cosine similarity + TAR@FAR=0.1% + AUC of ROC. Matcher is held out - not used in training or A7 selection.

* **FID:** Computed over >=5 000 reconstructed vs. ground-truth crops (not vs. training set). Measures distributional realism.

* **NIQE / BRISQUE:** No-reference per-image quality. Catches unrealistic outputs that may fool FID.

* **Landmark error:** Mean L2 distance (px) between predicted and GT landmarks after reconstruction - structural fidelity measure.

* **CLIP score:** Text-guided experiments only: cosine(CLIP(output), CLIP(prompt)). Evaluates attribute preservation (e.g. 'wearing glasses').

ABLATION CONDITIONS

- **A1** Single reference (random)
- **A2** Random multi-reference, no scoring
- **A3** Globally best reference (quality, no mask-aware)
- **A4** Mean multi-reference fusion (equal weights)
- **A5** Global attention fusion (no mask conditioning)
- **A6** Proposed mask-aware regional fusion, no verifier
- **A7** Proposed fusion + verifier-aware sampling [FULL]

*Held-out matcher: ArcFace-R100 (MS1MV3) - NOT used in training loss or A7 selection.*

*Verification: reconstructed face vs. >=3 withheld references per identity. Report cosine sim + TAR@FAR=0.1% + AUC.*

# Implementation Strategy — Three-Review Plan

**Review 1**
**(Month 1–3)**

## Retrieve Module

- Implement mask geometry analyser
- Build reference scoring pipeline (pose + quality + overlap)
- Validate scoring on CelebA-HQ with synthetic masks
- Baseline: single-reference ArcFace-conditioned inpainting

**Review 2**
**(Month 4–7)**

## Fuse Module

- Implement patch-level identity encoder
- Build mask-conditioned cross-attention fusion
- Integrate fused tokens into diffusion UNet conditioning
- Ablations A1–A5 completed

**Review 3**
**(Month 7–10)**

## Verify Module

- Build verifier-aware candidate selection (A7)
- Tune joint identity × quality scoring policy
- Run primary experiment across 4 mask-severity levels
- Evaluate with held-out matcher; compile final results

# Project Timeline — 10 Months

| M1–2 | M3 | M4–5 | M6–7 | M8 | M9 | M10 |
|---|---|---|---|---|---|---|
| Dataset Setup & Baselines | Mask-Aware Scoring | Identity Encoder & Fusion | Diffusion Integration & Ablations | Verifier-Aware Selection | Primary Experiment & Eval | Paper Writing & Submission |

**Review milestones: Review 1 → End of M3 | Review 2 → End of M7 | Review 3 → End of M10**

## DELIVERABLES PER REVIEW

**Review 1:** Scored reference retrieval + baseline comparison (A1 vs A3). Dataset preprocessing complete.

**Review 2:** Full Retrieve + Fuse modules. Ablations A1–A5 on CelebA-HQ at 50% mask severity.

**Review 3:** Complete proposed system (A7). Primary experiment across 4 severity levels. Held-out matcher evaluation. Draft paper.